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Prepared for

MUMINJON

ANARITA BESS

10 MW / 20–40 MWh Battery Energy Storage
System

Paphos District, Cyprus

TURNKEY EPC · BANKABLE GUARANTEES · LOCAL O&M

Technology Partners:



KEHUA

5 March 2026
Lighthief Cyprus Ltd

Executive Summary

Lighthief Cyprus Ltd is pleased to present this commercial offer to **Muminjon** for a fully bankable, turnkey Battery Energy Storage System (BESS) at the Anarita site in Paphos District. The system uses Linyang Power Atlantic LFP battery containers with Kehua C-series grid-following/grid-forming PCS inverters (TÜV-certified EN 50549-2). We offer three configurations to match your investment horizon and revenue strategy. All prices include equipment, installation, commissioning, and delivery to site (DAP Anarita).

Note: Pricing assumes standard ground conditions; site-specific ground works and levelling are excluded. DSO grid connection works, if required, are not included.

Option A 10 MW / 20 MWh	Option B 10 MW / 30 MWh	Option C RECOMMENDED 10 MW / 40 MWh
€3,033,000	€4,040,000	€5,048,000
€151.65 per kWh	€134.68 per kWh	€126.20 per kWh
Duration 2 hours	Duration 3 hours	Duration 4 hours
Battery 4 × 5 Containers MWh	Battery 6 × 5 Containers MWh	Battery 8 × 5 Containers MWh
PCS 8 × 1.25 Units MW	PCS 8 × 1.25 Units MW	PCS 8 × 1.25 Units MW
DeliveryDAP Anarita Site	DeliveryDAP Anarita Site	DeliveryDAP Anarita Site

Why Option C (10/40) is Recommended:

- Lowest cost per kWh (€126.20 vs €134.68 vs €151.65) — best value
- 4-hour duration maximizes revenue from energy time-shifting
- Better alignment with Cyprus electricity market peak periods
- Higher total energy capacity for grid services and ancillary revenue

What's Included

Component	Option A (20 MWh)	Option B (30 MWh)	Option C (40 MWh)
Battery Containers (Linyang Power Atlantic)	4 × 5 MWh (20 MWh)	6 × 5 MWh (30 MWh)	8 × 5 MWh (40 MWh)
PCS (Kehua BCS1250K-C-HUD)	8 × 1.25 MW (10 MW)	8 × 1.25 MW (10 MW)	8 × 1.25 MW (10 MW)
MV SKID (Transformer + Switchgear)	10 MW rated	10 MW rated	10 MW rated
Delivery	DAP Anarita Site (Turnkey)		
Installation & Commissioning	Full turnkey installation		
Base Warranty	5 years (equipment)		

Technical Specifications

Battery System - Linyang Power Atlantic

Parameter	Specification
Manufacturer	Jiangsu Linyang Energy Storage Technology Co., Ltd
Cell Chemistry	LFP (Lithium Iron Phosphate)
Cell Capacity	314 Ah (EVE)
Container Size	5.015 MWh per 20HC container
DC Voltage Range	1,164.8 - 1,497.6 V
Cycle Life	8,000 cycles @ 90% DOD, 70% EOL
Round-Trip Efficiency	≥87.8% (system level)
Operating Temperature	-30°C to +55°C (discharge)
Cooling System	Liquid cooling (45 kW capacity)
IP Rating	IP55 (container)
Fire Suppression	Aerosol (container + pack level)

Power Conversion System - Kehua BCS1250K-C-HUD (C-Series)

Parameter	Specification
Manufacturer	Xiamen Kehua Digital Energy Tech Co., Ltd

Rated Power 1,250 kW per unit (8 units = 10 MW)

Maximum Efficiency	≥99%
AC Voltage	690 VAC (3-phase)
Grid Frequency	50 Hz
Power Factor	> 0.99 @ rated power
THDi	< 3% @ rated power
Operating Modes	Grid-Following (default) & Grid-Forming (VSG, Black-Start, PQ, VF)
C-Series Designation	Latest-generation PCS with built-in grid-forming controller — no external add-on required
Response Time	Millisecond-level
IP Rating	IP65
Corrosion Protection	C5 rated

C-Series PCS — Grid-Following & Grid-Forming Ready

The Kehua BCS1250K-**C**-HUD is the latest-generation C-series inverter with a built-in grid-forming controller. In standard operation, the PCS runs in **grid-following mode**, synchronising to the existing grid signal. When required, it can seamlessly transition to **grid-forming mode** (Virtual Synchronous Generator), providing voltage and frequency support, black-start capability, and island operation. This dual-mode capability future-proofs the installation for upcoming Cyprus TSO grid code requirements and enhances ancillary service revenue potential. TÜV-certified to **EN 50549-2:2019/A1:2023** (certificate D 115067 0077).

State of Health (SOH) Guarantee

Year	1 Cycle/Day (0.25P)	2 Cycles/Day (0.25P)
Year 1	94.62%	93.91%
Year 5	86.78%	83.72%
Year 10	79.58%	74.13%
Year 15	73.61%	N/A (EOL @ Year 12)

Values based on Linyang degradation curves at 0.25P rate. EOL defined as 70% SOH.

Certifications

- UL 9540A
- IEC 62619
- IEC 63056
- EN 50549-2
- UN38.3
- EN 62477-1
- IEC 61000-6-2/6-4

Scope of Supply

✓ INCLUDED IN EPC PRICE

- **Battery Containers** - Complete with LFP cells, BMS, HVAC, fire suppression
- **Power Conversion Systems** - Kehua PCS units with grid-forming capability
- **MV SKID** - Transformer, switchgear, protection systems
- **Auxiliary Transformer** - For HVAC and BMS power
- **RTU** - Remote Terminal Unit for DSO communication
- **Shipping & Transport** - CIF Limassol + inland delivery to site
- **Import Duties & Customs** - Clearance and handling
- **Inland Transport** - Port to site delivery
- **Concrete Foundations** - Reinforced per K.Δ.Π. 15/2026 (assumes level ground*)
- **DC Cabling** - Up to 50m per container
- **AC/MV Cabling** - LV and MV connections
- **MV Terminations** - Connection to grid
- **Protection Engineering** - Settings and coordination
- **SCADA Programming** - IEC 60870-5-104 integration
- **Remote Trip System** - DSO emergency disconnect
- **Commissioning** - Full system commissioning
- **Insurance** - CAR/EAR during

✗ CLIENT RESPONSIBILITY

- **CERA License** - Generation license application
- **DSO Application** - EAC connection agreement
- **DSO/EAC Fees** - Connection fees, network charges
- **External Lightning Protection** - If required by site assessment
- **Telecoms Backhaul** - Internet/4G connectivity (OPEX)
- **Site Fencing** - Perimeter security
- **Planning Permits** - If required for site
- **Land** - Site lease or purchase
- **Grid Upgrades** - If required by DSO
- **Water Drainage** - If special works required (€1,500)
- **Ground Works & Leveling** - Beyond standard scope (clearing, grading, excavation)
- **Access Roads** - If not existing or requiring upgrade

construction

- **OEM Supervision** - Linyang/Kehua engineer support
- **Base Warranty** - 5 years equipment warranty

* **Ground Conditions Assumption:** Pricing assumes reasonably level, stable ground suitable for standard foundations. Any site-specific ground works including but not limited to: land clearing, vegetation removal, grading, leveling, rock breaking, soil stabilization, or special drainage are **not included** and will be quoted separately upon site survey if required.

Project Timeline



Phase	Duration	Notes
Contract Signing & Advance Payment	Week 0	25% advance triggers production
Design Lock & Approval	Weeks 1-2	Final design signed by client
Civil Works (Parallel)	Weeks 2-12	Foundations can start during manufacturing
Manufacturing	90 days	Battery containers, PCS, MV equipment
Shipping & Delivery to Site	~50 days indicative	Sea freight + inland transport to Anarita
Installation & Commissioning	30 working days	From site delivery
Total to Commercial Operation	~6 months	From contract signing

Note: Civil works (foundations) can commence during the manufacturing phase, reducing overall project timeline. This parallel track provides 7+ weeks buffer before BESS arrival.

Long-Term Service Agreement (LTSA)

Lighthief Cyprus operates a permanent local service team to maintain and monitor your BESS throughout its operational life. All rates below are OEM-confirmed (Linyang RFI V3, February 2026) and contractually fixed at signing.

Service Detail	Provision
Service Provider	Lighthief Cyprus Ltd (local team, 6 field engineers)
Critical Alarm Response	4 hours remote / 8 hours on-site
Monitoring	24/7/365 via Voltus SCADA + Linyang Cloud EMS
Spare Parts	Local warehouse (Limassol) — critical spares stocked
Contract Duration	Up to 15 years (renewable in 5-year blocks)

Tier C: Preventive & Corrective Maintenance

Service	Rate (€/MWh/Year)	Option A (20 MWh/Year)	Option B (30 MWh/Year)	Option C (40 MWh/Year)
BESS Maintenance	€1,157.62	€23,152	€34,729	€46,305
PCS + MV Maintenance	€1,311.97	€26,239	€39,359	€52,479
Total Annual Maintenance	€2,469.59	€49,392/year	€74,088/year	€98,784/year

Included in Tier C:

- 24/7/365 remote monitoring via Voltus SCADA and Linyang Cloud EMS
- Bi-annual on-site preventive maintenance visits by certified engineers
- Corrective maintenance response (4-hour remote / 8-hour on-site for critical alarms)
- Software updates, firmware patches, and BMS parameter optimisation
- Monthly performance reporting and degradation tracking

- PCS inverter diagnostics and grid-forming controller calibration
- HVAC and liquid cooling system servicing

Extended Warranty Options

Component	Years 1-5 (Included)	Years 6-10	Years 11-15	Years 16-20
BESS Extended Warranty	✓ Included	+€913.92/MWh	+€1,157.62/MWh	+€3,858.75/MWh
PCS+MV Extended Warranty	✓ Included	+€747.76/MWh	+€926.10/MWh	+€2,315.25/MWh
Total Warranty Extension	€0	+ €1,661.68/MWh	+ €2,083.72/MWh	+ €6,174.00/MWh

20-Year Cost Summary (Option C: 40 MWh)

Years 1-5	Years 6-10	Years 11-15	Years 16-20
€2,469.59	€4,131.27	€4,553.31	€8,643.59
€98,784/year	€165,251/year	€182,132/year	€345,744/year

Tier C: Availability Guarantee (Included in Premium)

Service	Rate	Annual Cost (40 MWh)
97% Availability Guarantee	€2,201.73/MWh/Year	€88,069/year

Tier C Includes:

- Local service team in Cyprus
- Local spare parts warehouse
- Guaranteed 97% system availability
- Liquidated damages for availability shortfall

Note: Tier C is the highest service tier offered.

Commercial Terms

Payment Schedule

Milestone	Percentage	Option A (20 MWh)	Option B (30 MWh)	Option C (40 MWh)
Advance Payment (Contract Signing)	30%	€909,900	€1,212,150	€1,514,400
Ready to Ship (Production Complete)	50%	€1,516,500	€2,020,250	€2,524,000
Arrival on Site	15%	€454,950	€606,075	€757,200
SAT (Site Acceptance Test)	5%	€151,650	€202,025	€252,400
TOTAL	100%	€3,033,000	€4,040,500	€5,048,000

Performance Guarantees

Guarantee	Coverage
Installation & Commissioning Delay LDs	0.1-0.2% per day, capped at 10% (excludes shipping, DSO delays, Force Majeure)
Performance Bond	Corporate guarantee from Linyang Energy
SOH Guarantee	Per degradation curve, remedy via augmentation or pro-rata compensation
Availability (Tier C)	97% guaranteed with LDs for shortfall

Bankability & Investor Protection

This project is structured to meet the requirements of project financiers, lending institutions, and institutional investors. All contractual protections use fixed, auditable formulae with no discretionary elements.

Protection	Detail
Performance Bond	5% corporate bank guarantee from Linyang Energy (Shanghai-listed, revenue >US\$3B)
Delay Liquidated Damages	Fixed rate: 0.1–0.2% of contract value per day, capped at 10%. Accrual starts after agreed milestone dates (shipping & DSO delays excluded)
SOH Liquidated Damages	If SOH falls below guaranteed curve: shortfall MWh × €30/MWh/day. 90-day remedy grace period before LD accrual
Availability Guarantee	97% annual system availability. Shortfall compensated via defined LD mechanism
EPC Insurance	CAR/EAR + Third Party Liability during construction. Marine cargo covered by OEM (CIF terms)
OEM Certifications	TÜV-certified EN 50549-2 grid code compliance; IEC 62619 battery safety; UN38.3 transport certification
Equipment Warranty	5-year base warranty (included). Extendable to 15 years via OEM contract at confirmed rates
Local O&M Presence	Lighthief Cyprus service team (6 field engineers), spare parts warehouse in Limassol, 24/7 remote monitoring via Voltus SCADA

End-of-Life Management

Included at no additional cost: At end of BESS operational life, Lighthief Cyprus Ltd will:

- Collect and transport batteries at no cost to client
- Assess residual value for second-life or recycling
- Provide cashback based on EOL market value, OR
- Recycle at net zero cost to client

Offer Validity: 30 Days

This offer is valid until **4 April 2026**. Prices reflect manufacturer adjustments effective March 2026 and cannot be guaranteed beyond this date.

Acceptance

By signing below, **Muminjon** confirms acceptance of [] **Option A (10 MW / 20 MWh)**, [] **Option B (10 MW / 30 MWh)**, or [] **Option C (10 MW / 40 MWh)** (please check one) under the terms and conditions set forth in this offer.

FOR AND ON BEHALF OF MUMINJON:

FOR AND ON BEHALF OF
LIGHTHIEF CYPRUS LTD:

Name / Title / Date

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Cyprus Director

Contact Information

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